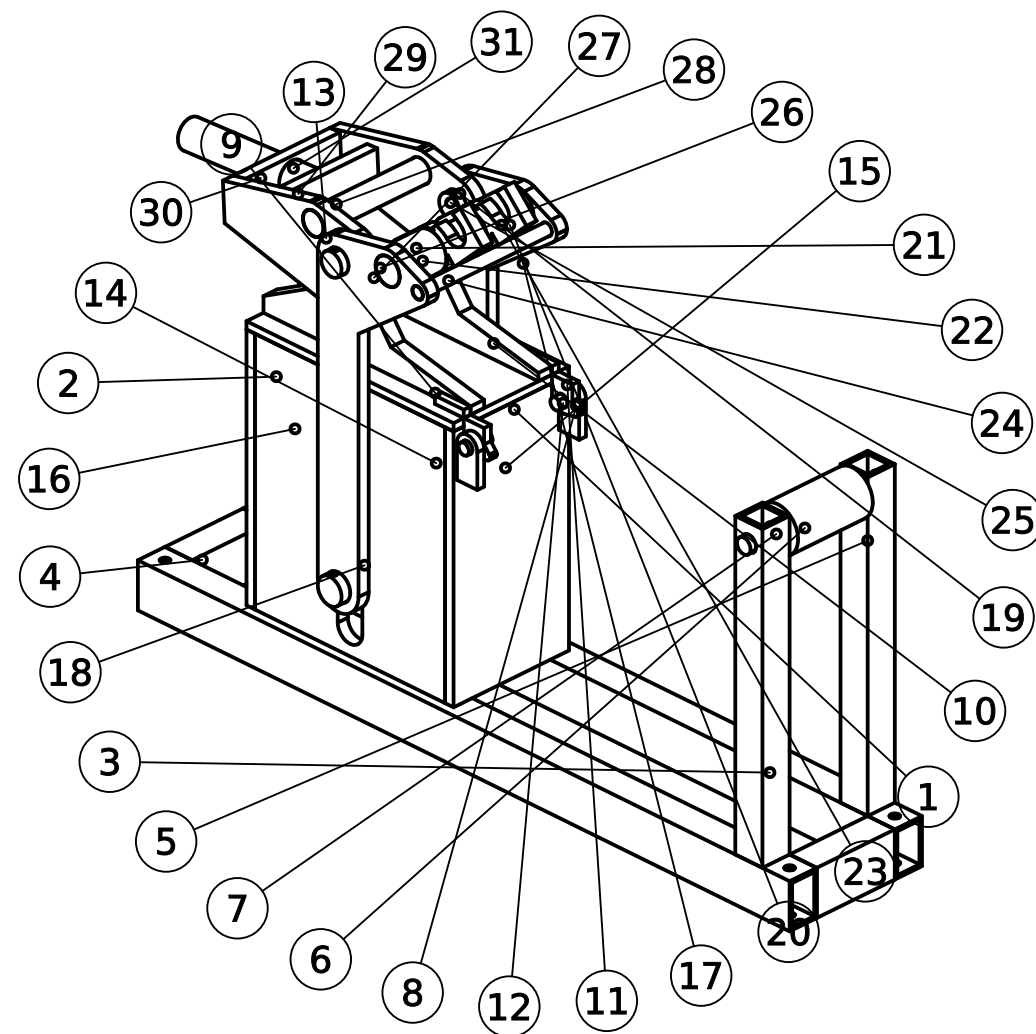
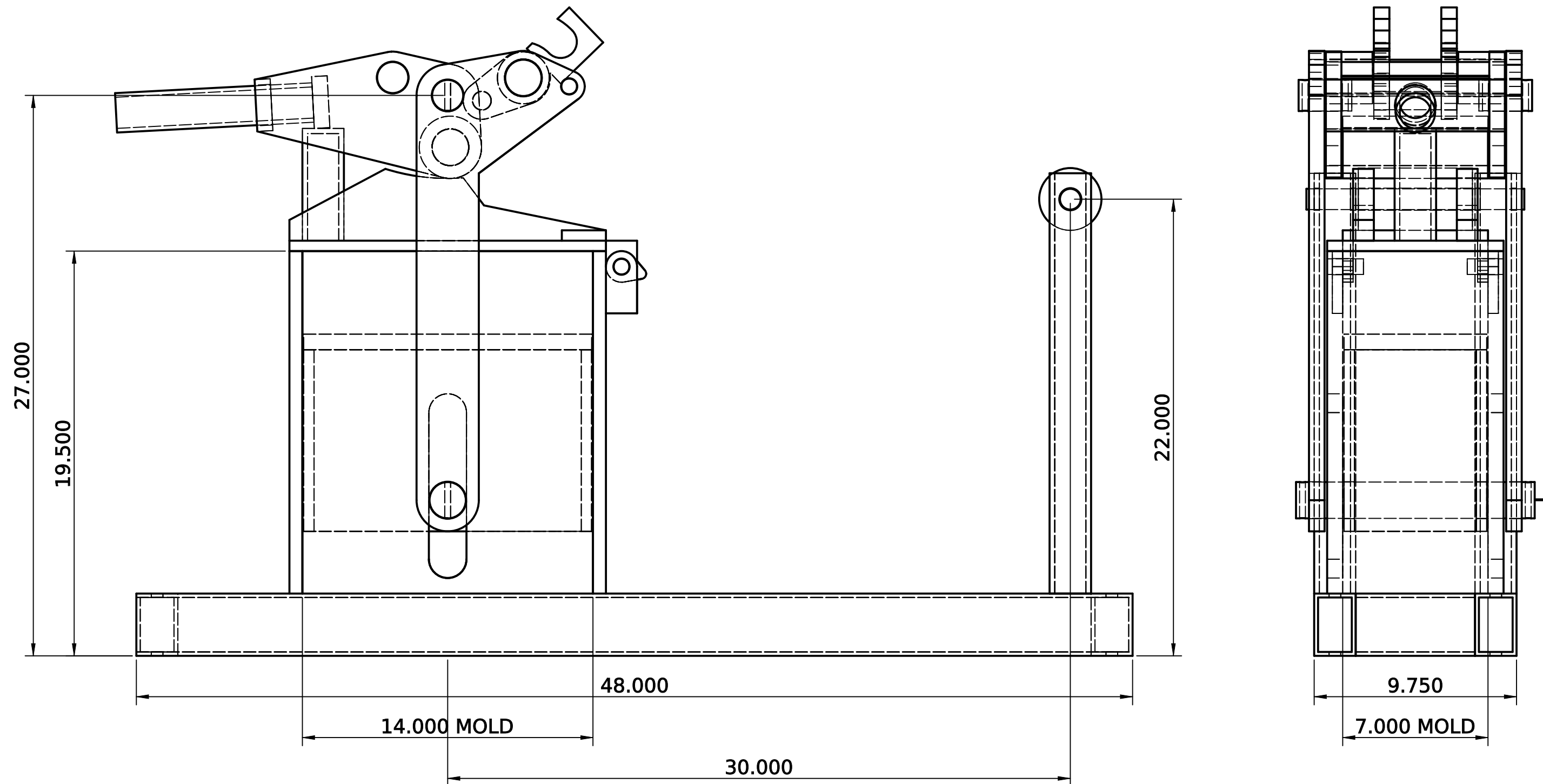




ISOMETRIC VIEW

- NOTES:
- 1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 - 2. BREAK SHARP EDGES, REMOVE BURRS.
 - 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
 - 4. BRICK 14 X 7 X 4; FILL 6.75 DEEP (RATIO 1.69).
 - 5. SHOWN COMPRESSED AND LOCKED. HANDLE SHOWN BROKEN, FULL LENGTH 72.
 - 6. SEE SHEET 4 FOR OPERATION, LOADS AND STRENGTH CHECKS.
 - 7. STEEL WEIGHT ABOUT 459 LB.

31	1	Handle Tube	1-1/2" sch 80 pipe	19.8	CR-14x7x4-31
30	1	Handle Bridge, Bored	0.750" plate A36	3.1	CR-14x7x4-30
29	1	Handle Bridge	0.750" plate A36	3.8	CR-14x7x4-29
28	1	Catch Bar	1.500" round 4140 prehard	4.3	CR-14x7x4-28
27	1	Roller Axle	1.750" round 4140 prehard	5.9	CR-14x7x4-27
26	1	Saddle Roller	3" OD 4140, 40-45 HRC	9.1	CR-14x7x4-26
25	2	Handle Cheek	0.750" plate A36	9.0	CR-14x7x4-25
24	1	Claw Stop Rod	0.750" round 1018	1.3	CR-14x7x4-24
23	2	Tie Spacer	2.5" OD round 1018	1.5	CR-14x7x4-23
22	1	Claw Tie	1.750" round 4140 prehard	7.0	CR-14x7x4-22
21	1	Claw Thumb Bar	0.875" round 1018	0.7	CR-14x7x4-21
20	2	Claw Plate	0.750" plate A36	2.1	CR-14x7x4-20
19	2	Stub Pin Q	1.500" round 4140 prehard	1.2	CR-14x7x4-19
18	1	Main Pin P	1.750" round 4140 prehard	7.8	CR-14x7x4-18
17	2	Yoke Bar	0.750" plate A36 (profile)	16.4	CR-14x7x4-17
16	2	Piston Diaphragm	0.500" plate A36	7.3	CR-14x7x4-16
15	2	Piston Web	0.500" plate A36	16.9	CR-14x7x4-15
14	1	Piston Cap	0.750" plate A36	20.3	CR-14x7x4-14
13	1	Handle Stop	2x2x1/4 sq. tube + cap	2.8	CR-14x7x4-13
12	2	Hinge Pin	0.750" round 1018	0.2	CR-14x7x4-12
11	2	Lid Hinge Lug	0.500" plate A36	0.5	CR-14x7x4-11
10	2	Lid Rib (Roller Track)	1.000" plate AR400	8.8	CR-14x7x4-10
9	1	Lid Plate	0.500" plate A36	18.4	CR-14x7x4-09
8	2	Wall Hinge Lug	0.500" plate A36	0.5	CR-14x7x4-08
7	1	Eject Roller Axle	1.000" round 1018	2.3	CR-14x7x4-07
6	1	Eject Roller	3" OD round, 1045	9.6	CR-14x7x4-06
5	2	Eject Roller Post	2 x 2 x 1/4 sq. tube	9.9	CR-14x7x4-05
4	2	Cross Tie	3 x 2 x 3/16 rect. tube	2.8	CR-14x7x4-04
3	2	Base Rail	3 x 2 x 3/16 rect. tube	23.6	CR-14x7x4-03
2	2	End Wall	0.625" plate A36	24.9	CR-14x7x4-02
1	2	Side Wall	0.750" plate A36	45.9	CR-14x7x4-01
Item	Qty	Title	Stock	lb ea	Drawing no.
Material		Qty	Scale	Units	Projection
See parts list		1	1:10	inch	
Created by		Approved by	Document type		Status
J. Clements III		-	Assembly drawing		Draft
Legal owner		Title		Identification number	Sheet
J. Clements III		CINVA-Ram Block Press		CR-14x7x4-00	1/35
CEB Project		General assembly, 14x7x4 brick		Rev A Date of issue 2026-09-30	



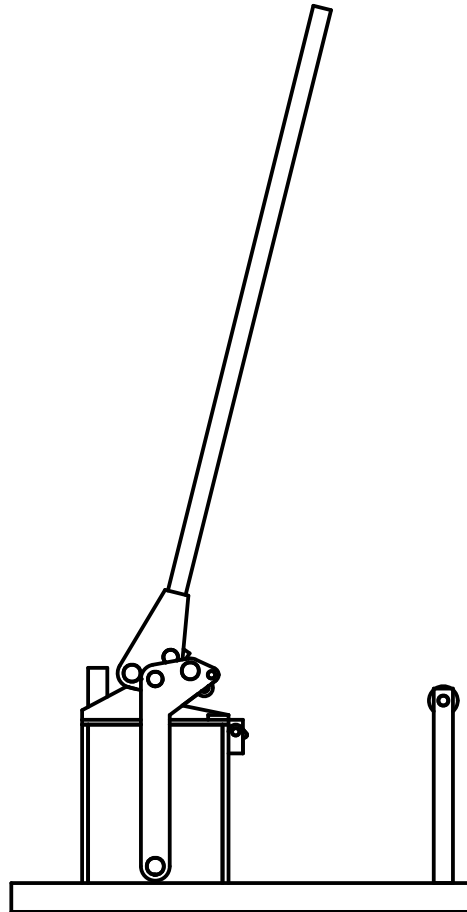
NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. BRICK 14 X 7 X 4; FILL 6.75 DEEP (RATIO 1.69).
5. SHOWN COMPRESSED AND LOCKED. HANDLE SHOWN BROKEN, FULL LENGTH 72.
6. SEE SHEET 4 FOR OPERATION, LOADS AND STRENGTH CHECKS.

Material See parts list		Qty 1	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Assembly drawing	
Legal owner J. Clements III CEB Project		Title CINVA-Ram Block Press Assembly views, 14x7x4 brick		Identification number CR-14x7x4-00	Sheet 2/35
				Rev A	Date of issue 2026-09-30

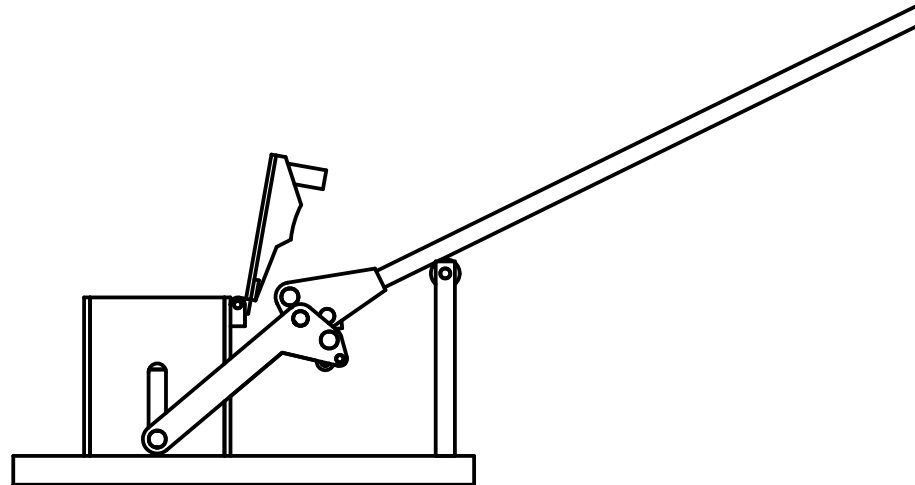
2 START OF COMPRESSION

yoke 0.0 deg, handle -14.0 deg, piston -6.750



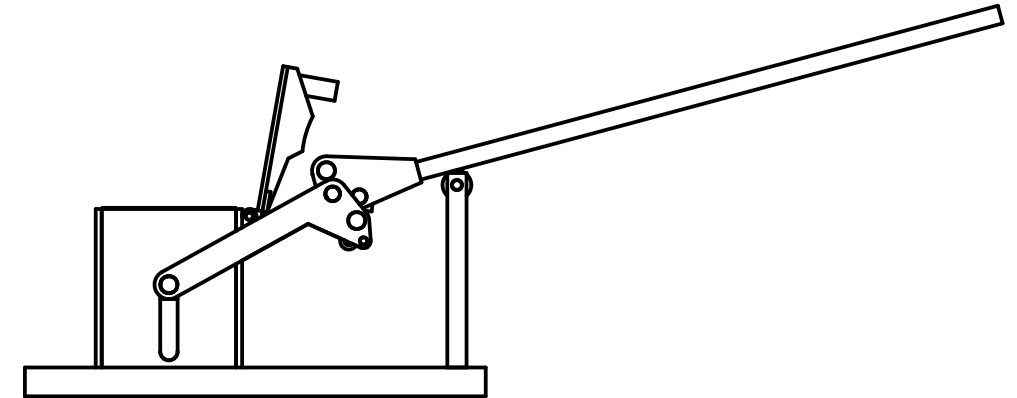
1 FILL

yoke 49.9 deg, handle -14.0 deg, piston -6.750



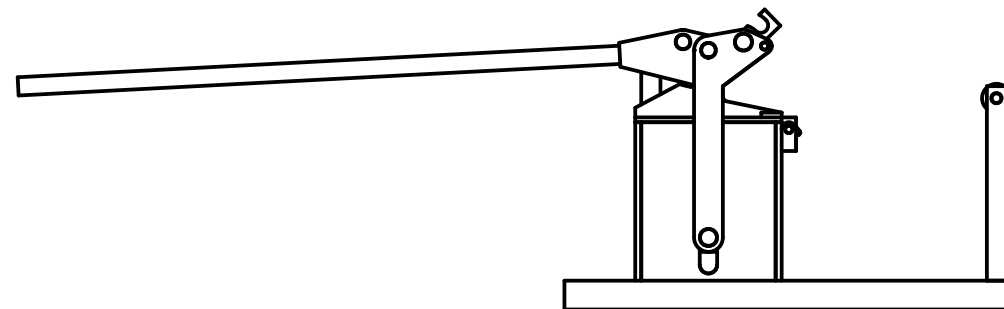
4 EJECT

yoke 61.0 deg, handle -14.0 deg, piston +0.125



3 COMPRESSED AND LOCKED

yoke 0.0 deg, handle 93.0 deg, piston -4.002



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. POSITIONS COMPUTED FROM THE LINKAGE; ALL FOUR AND THE MOTIONS BETWEEN THEM WERE CHECKED FOR CLEARANCE.
5. YOKE TILT FROM VERTICAL TOWARD THE EJECT ROLLER; HANDLE ANGLE FROM THE YOKE AXIS, POSITIVE AWAY FROM THE EJECT ROLLER; PISTON = PISTON TOP RELATIVE TO THE MOLD TOP.

Material	-	Qty	1	Scale	1:20	Units	inch	Projection	
Created by	J. Clements III	Approved by	-	Document type	Assembly drawing	Status	Draft		
Legal owner	J. Clements III CEB Project	Title	CINVA-Ram Block Press Operating positions, 14x7x4 brick			Identification number	CR-14x7x4-00	Sheet	3/35
Rev	A	Date of issue	2026-09-30						

PRESS DATA

Brick (pressed)	14 x 7 x 4 in, 98 sq in face
Loose fill depth	6.750 in (strike off level with mold top)
Compression	2.750 in piston stroke, ratio 1.69
Working pressure	200 psi = 19,600 lbf on the piston
Design (overfill) load	306 psi = 29,988 lbf, all parts checked
Handle	72 in from pivot Q, swings -14 to 93 deg (over centre)
Peak push on handle	122 / 162 / 203 lbf at 150 / 200 / 250 psi
Ejection	piston rises 4.125 in; yoke tilts 54 to 61 deg
Steel weight	about 459 lb

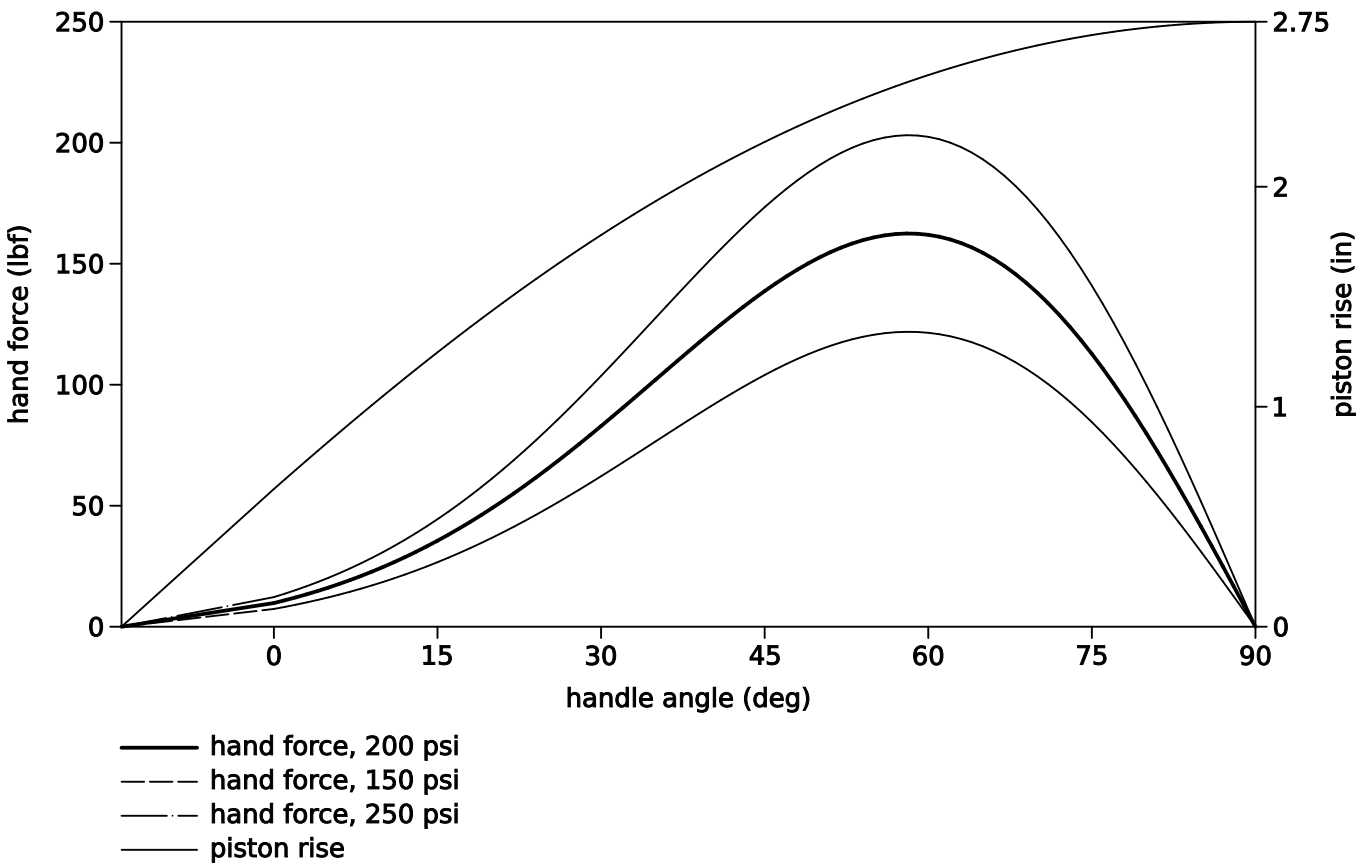
OPERATION

1. Latch claw shut on the catch bar (handle locked to the yoke), yoke tilted back onto the eject roller. Piston at bottom.
2. Open the lid. Oil the mold walls. Fill loose soil mix to the top, press into the corners, strike off level.
3. Close the lid. Swing handle and yoke upright together until the saddle roller sits on the lid track.
4. Push the claw thumb bar down to flip the claw open onto its stop rod. Pull the handle over, away from the eject roller, past horizontal onto the handle stop; it goes slightly over centre and stays there. Use two or three firm pushes; it must reach the stop.
5. Lift the handle back up over centre (it takes little force) to upright, drop the claw onto the catch bar, and tilt handle and yoke back onto the eject roller.
6. Open the lid toward the eject roller. Push the handle down to raise the brick clear of the mold. Lift the brick straight off by its ends.
7. Weigh or measure each fill: too little soil gives weak bricks; too much overloads the press.

STRENGTH CHECKS AT 29,988 LBF (ksi)

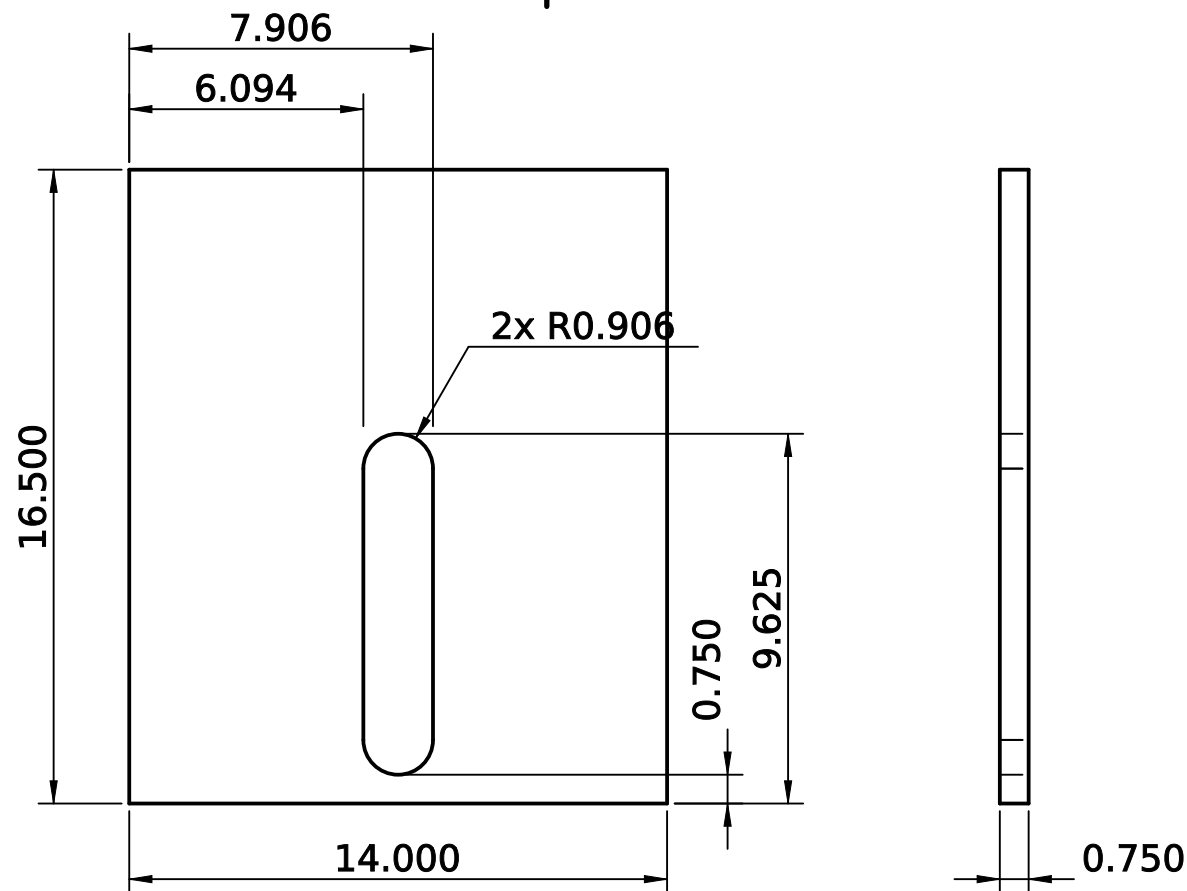
Item	Basis	Stress	Allow	Ratio
Main pin P, bending	$M = F/2 \times 1.56$, dia 1.750 4140	44.5	57.0	0.78
Main pin P, shear	F/2 per side	6.2	38.0	0.16
Stub pins Q, bending	$M = F/2 \times 0.81$, dia 1.500 4140	36.8	57.0	0.65
Roller axle, bending	$M = F/2 \times 1.44$, dia 1.750 4140	41.0	57.0	0.72
Yoke bar, net tension	F/2 on 0.91 sq in	16.4	29.0	0.57
Yoke bar, pin bearing	F/2 on 1.75 x 0.75	11.4	32.4	0.35
Cheek, Q pin bearing	F/2 on 1.50 x 0.75	13.3	32.4	0.41
Side wall, bending	K0 0.5, strip 6.88 x 0.750, ends fixed	15.5	23.8	0.65
End wall, bending	K0 0.5, band 4 x 0.625	14.1	23.8	0.60
Lid, bending at centre	$M = F L / 8$, $I = 12.2 \text{ in}^4$	9.6	23.8	0.40
Piston cap, bending	strip span 5.88, $t = 0.750$	14.1	23.8	0.59
Piston web, bending	$M = F L / 16$, net at pin hole	7.3	23.8	0.31
Claw tie, bending (eject)	12,245 lbf, 6,122 per plate x 3.12, dia 1.750 4140	56.4	57.0	0.64
Catch bar, bending (eject)	6,122 lbf per plate x 2.31, dia 1.500 4140	42.7	57.0	0.75
Claw notch, bearing (eject)	6,122 lbf on 1.50 x 0.75	5.4	32.4	0.17
Claw arm, tension (eject)	6,122 lbf on 1.50 x 0.75	5.4	21.6	0.25
Yoke horn, net at tie hole (eject)	6,122 lbf on (2.60 - 1.75) x 0.75	10.0	29.0	0.34
Roller on track at lock, contact	R 1.500 in concave R 10, AR400 rib	211.1	241.7	0.87

HAND FORCE AND PISTON RISE OVER THE STROKE



- NOTES:
1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 2. BREAK SHARP EDGES, REMOVE BURRS.
 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
 4. FIRST-ORDER HAND CALCULATIONS; BUILD ONE PRESS AND LOAD-TEST BEFORE PRODUCTION.
 5. ALLOWABLES: A36 0.66 FY BENDING, 0.5 FU NET TENSION; 4140 PREHARD 0.6 FY; AR400 CONTACT 0.3 P $\leq$ 0.5 FY.
 6. SOIL MODEL: PRESSURE RISES AS $(E^5 X - 1)/(E^5 - 1)$ OVER THE STROKE; LATERAL WALL PRESSURE 0.5 X VERTICAL.

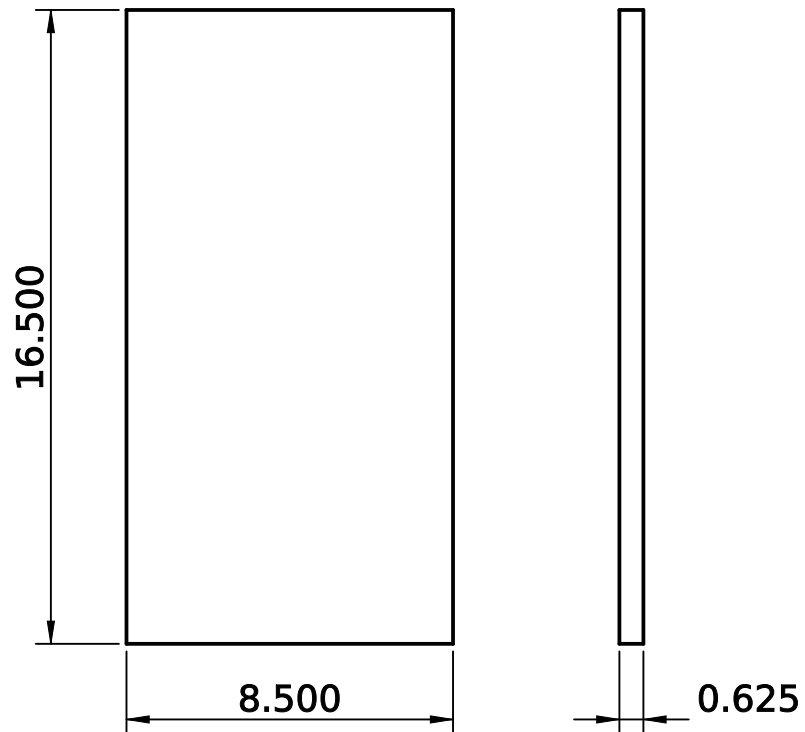
Material	Qty	Scale	Units	Projection
-	1	-	inch	
Created by	Approved by	Document type	Status	
J. Clements III	-	Design data	Draft	
Legal owner	Title	Identification number	Sheet	
J. Clements III	CINVA-Ram Block Press	CR-14x7x4-00	4/35	
CEB Project	Operation and design data, 14x7x4 brick	Rev A	Date of issue	2026-09-30



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. SLOT GUIDES MAIN PIN P; KEEP SLOT EDGES SMOOTH AND PARALLEL.
5. WELD TO END WALLS OUTSIDE ONLY; NO WELD OR SPATTER INSIDE THE MOLD.

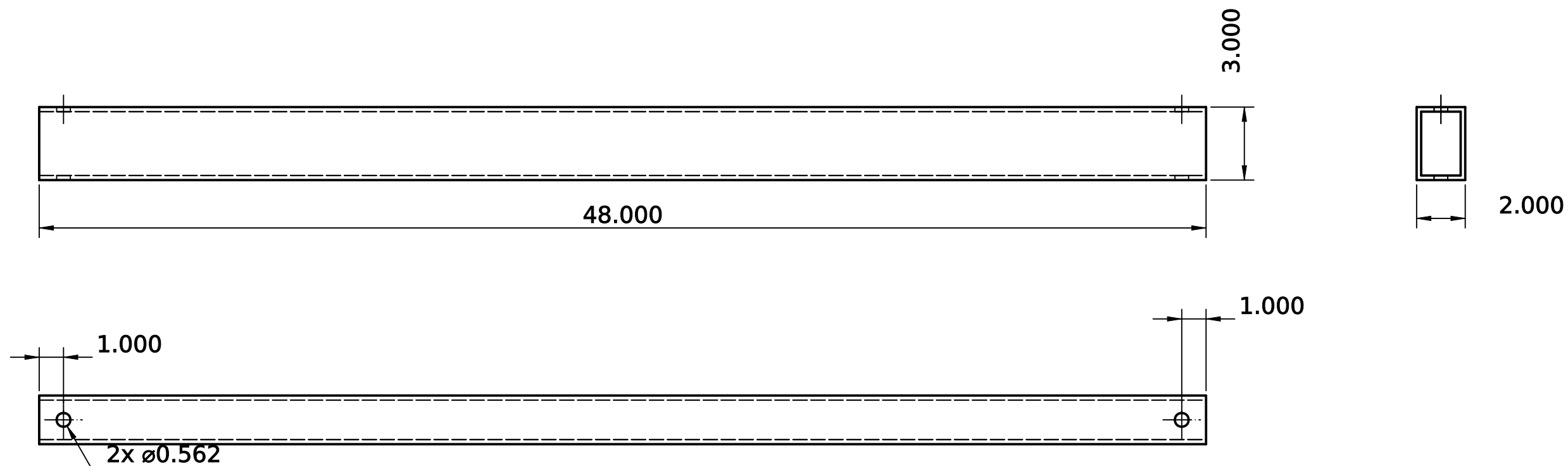
Material 0.750" plate A36		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Side Wall CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-01	Sheet 5/35
				Rev A	



NOTES:

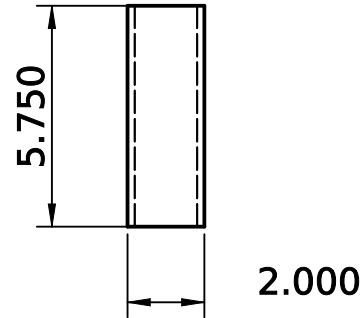
1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. MOLD INSIDE MUST BE SQUARE: CHECK DIAGONALS BEFORE FINAL WELDS.

Material 0.625" plate A36		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title End Wall CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-02	Sheet 6/35
				Rev A	



- NOTES:
1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 2. BREAK SHARP EDGES, REMOVE BURRS.
 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
 4. MOUNTING HOLES THROUGH BOTH WALLS, FOR 1/2 ANCHOR BOLTS OR LAGS.

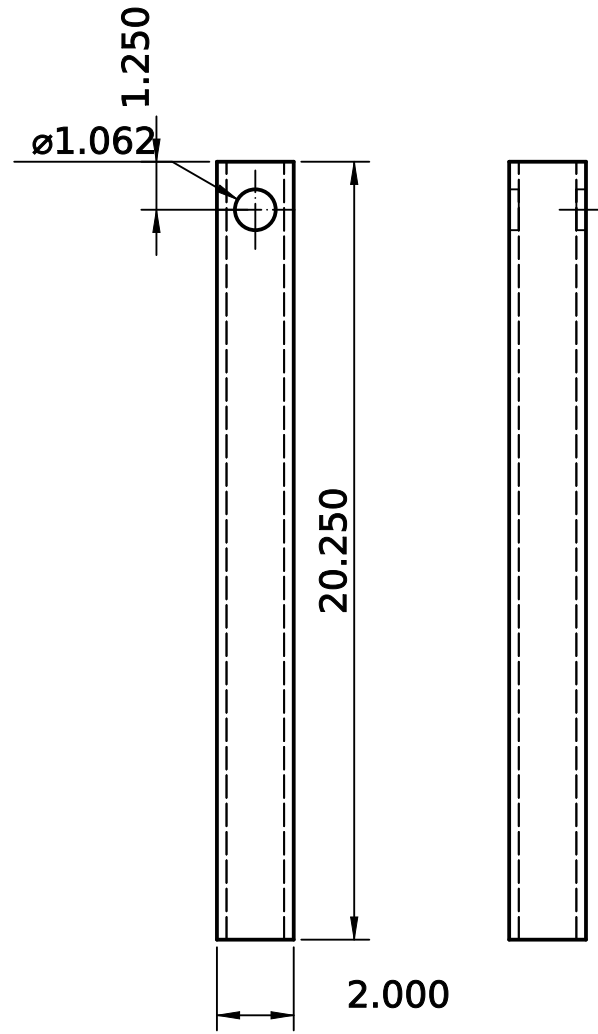
Material 3 x 2 x 3/16 rect. tube		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project		Title Base Rail CINVA-Ram, 14x7x4 brick		Identification number CR-14x7x4-03	Sheet 7/35
				Rev A	
				Date of issue 2026-09-30	



NOTES:

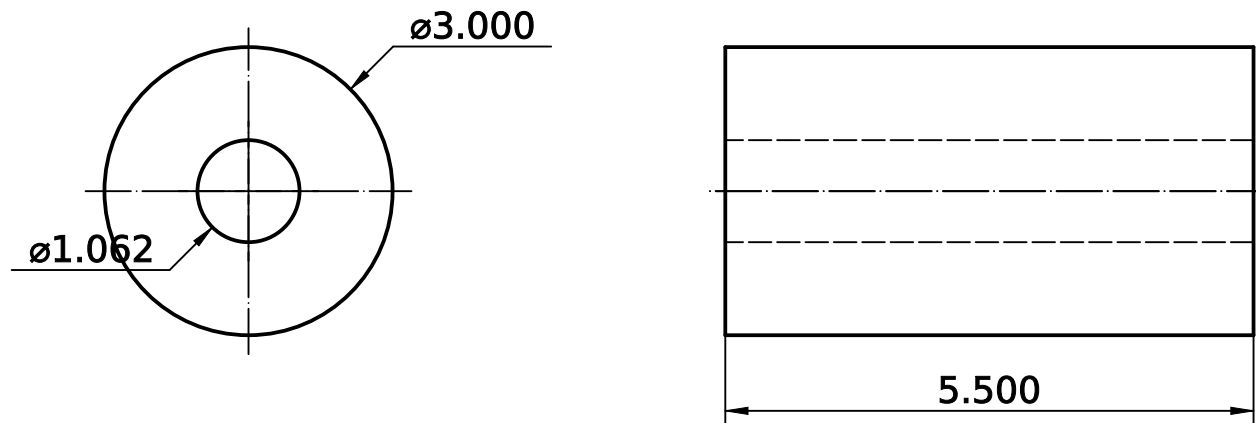
1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.

Material 3 x 2 x 3/16 rect. tube		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Cross Tie CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-04	Sheet 8/35
				Rev A	



- NOTES:
1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
 2. BREAK SHARP EDGES, REMOVE BURRS.
 3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.

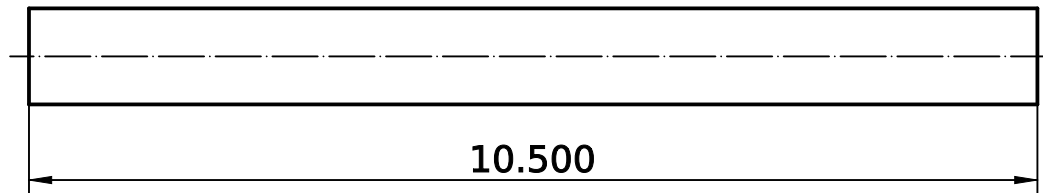
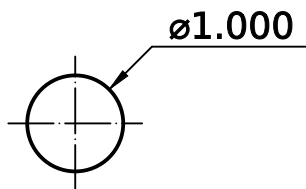
Material 2 x 2 x 1/4 sq. tube		Qty 2	Scale 1:5	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Eject Roller Post CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-05		Sheet 9/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.

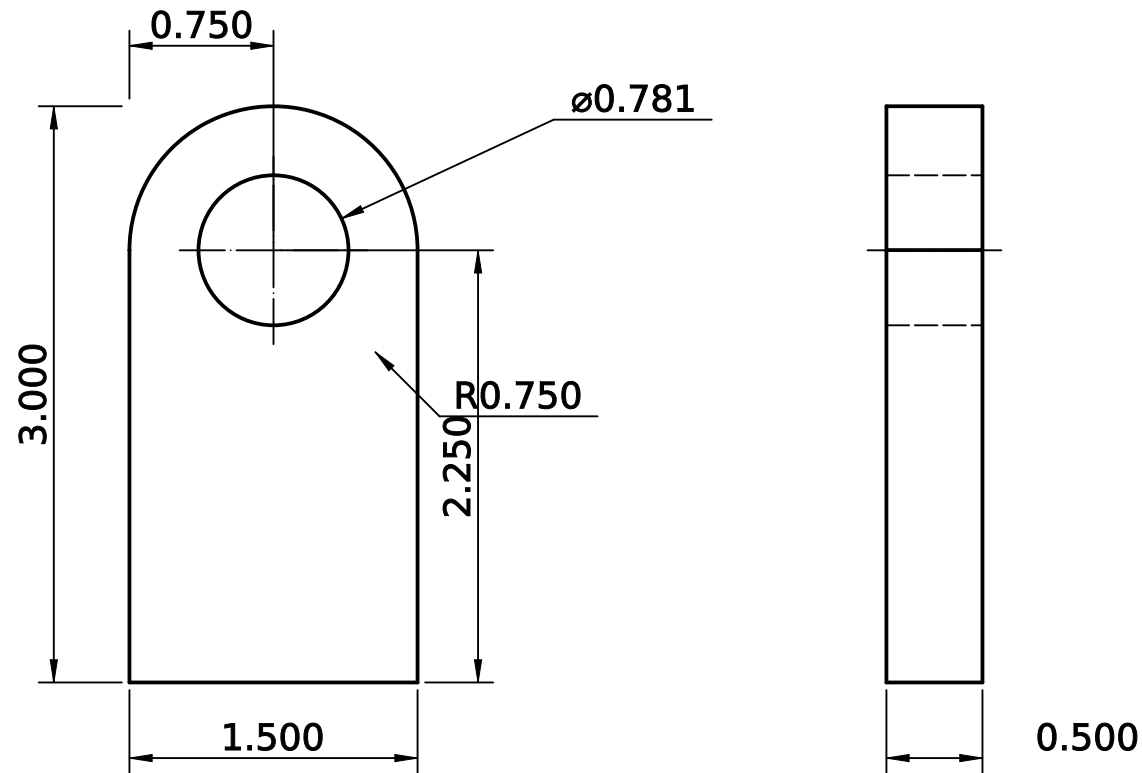
Material 3" OD round, 1045		Qty 1	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Eject Roller CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-06	Sheet 10/35
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.

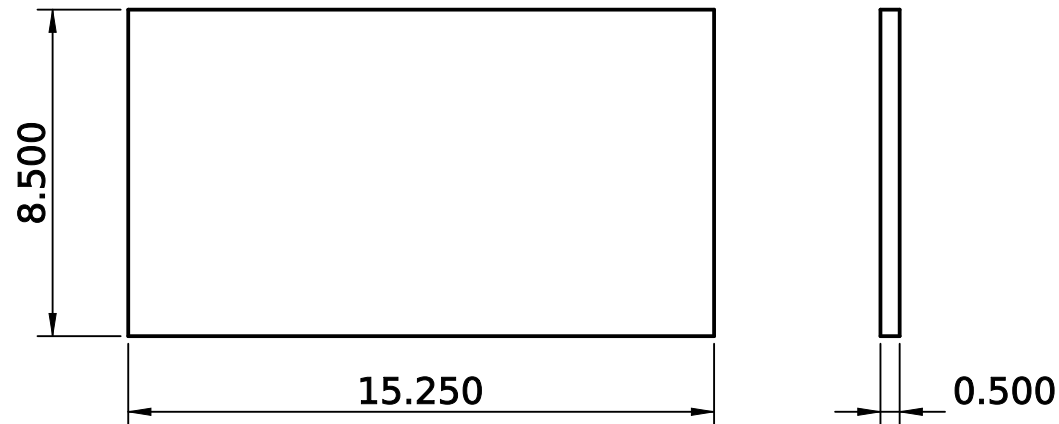
Material 1.000" round 1018		Qty 1	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Eject Roller Axle CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-07		Sheet 11/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. WELD TO THE OUTER FACE OF THE EJECT-END WALL, TOP FLUSH WITH THE MOLD TOP, OUTER FACE 0.250 IN FROM THE SIDE-WALL OUTER FACE.

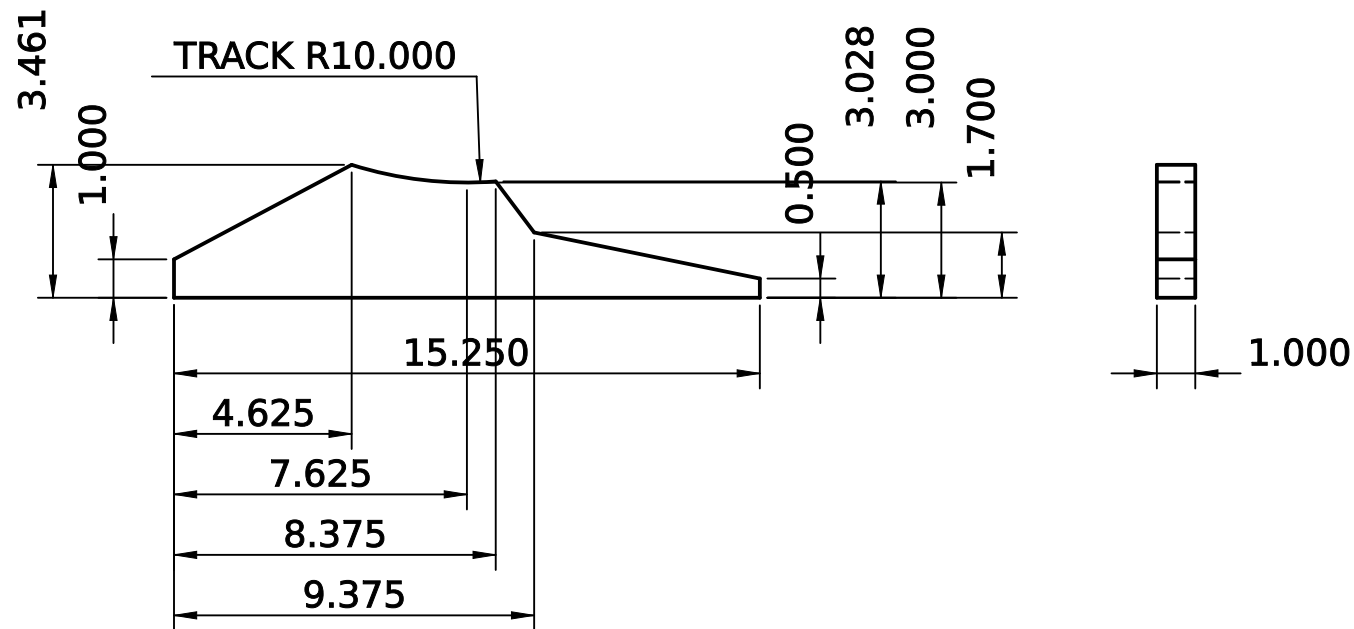
Material 0.500" plate A36		Qty 2	Scale 1:1	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Wall Hinge Lug CINVA-Ram, 14x7x4 brick	Identification number CR-14x7x4-08			Sheet 12/35
		Rev A	Date of issue 2026-09-30		



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. UNDERSIDE MUST BE FLAT AND CLEAN; IT FORMS THE TOP FACE OF THE BRICK.

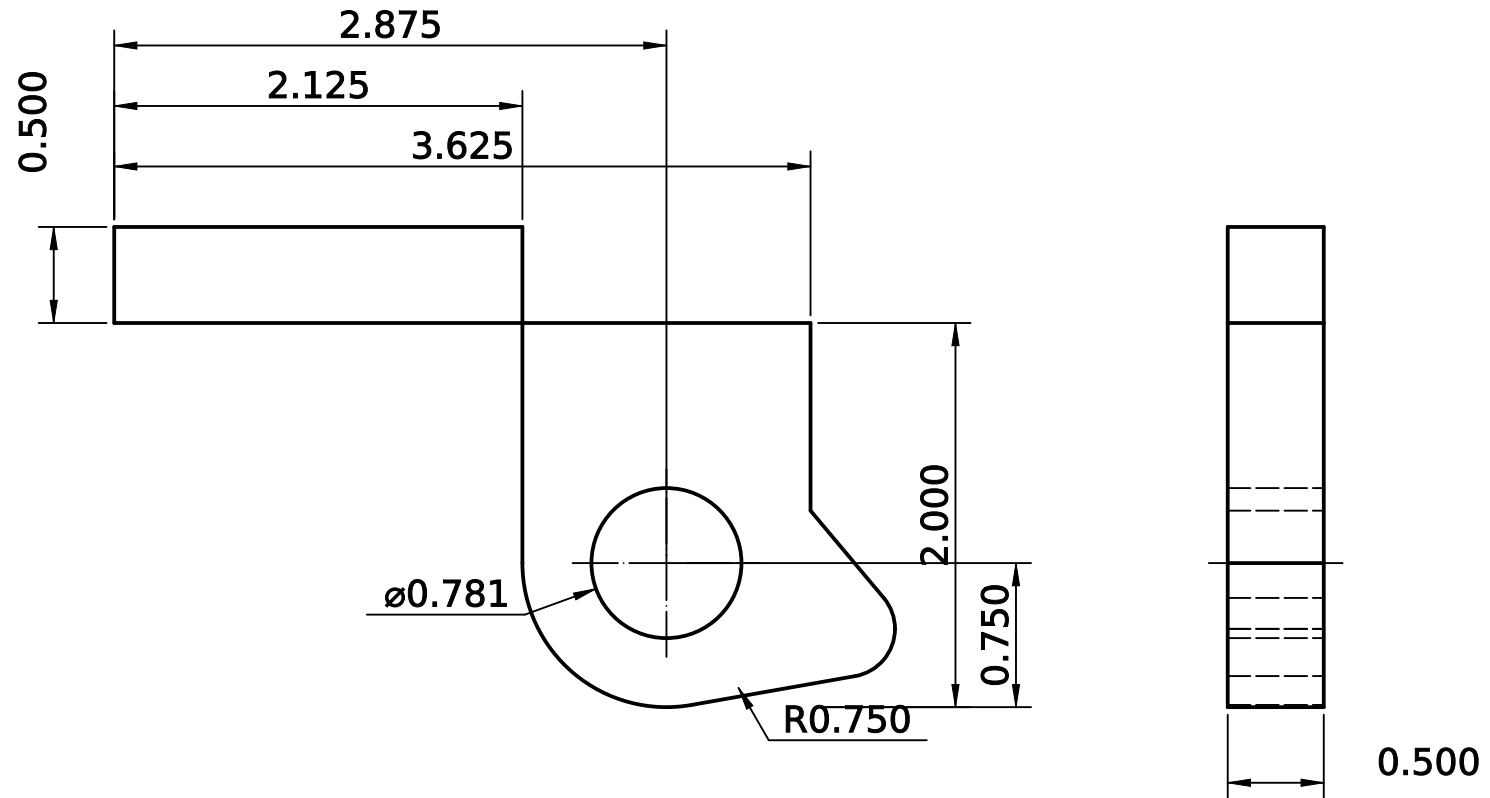
Material 0.500" plate A36		Qty 1	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Lid Plate CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-09	Sheet 13/35
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. AR400 WEAR PLATE: THE ROLLER BEARS ON THIS EDGE AT UP TO 15,000 LBF.
5. TRACK IS ONE CONCAVE ARC R10.000, LOWEST POINT ON THE MOLD CENTRELINE, 3.000 ABOVE THE LID; CUT FROM THE DXF, NO FLATS OR CORNERS BETWEEN THE TWO SLOPES.
6. GRIND THE TRACK SMOOTH AND SQUARE TO THE PLATE WITHIN 1/64.
7. WELD TO LID PLATE BOTH SIDES FULL LENGTH, LOW-HYDROGEN ROD, PREHEAT 300 F.

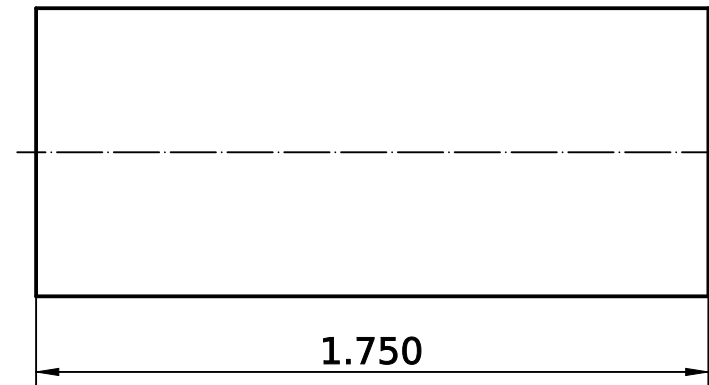
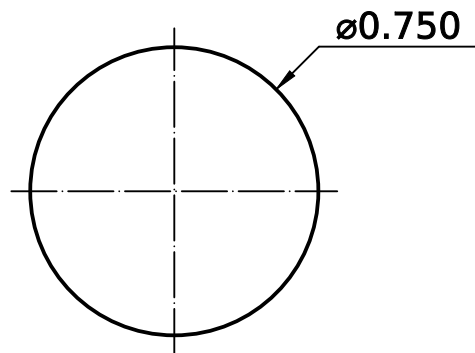
Material 1.000" plate AR400		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Lid Rib (Roller Track) CINVA-Ram, 14x7x4 brick	Identification number CR-14x7x4-10			Sheet 14/35
		Rev A	Date of issue 2026-09-30		



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. WELD ON THE LID TOP AT THE EJECT END, INNER EDGE AGAINST THE LID END FACE, OUTER FACE 0.750 IN FROM THE LID EDGE.
5. HINGE HOLE SITS BELOW THE LID SO THE LUG STAYS CLEAR OF THE SADDLE ROLLER.
6. THE TAIL LOBE RESTS ON THE END WALL WITH THE LID OPEN JUST PAST VERTICAL.

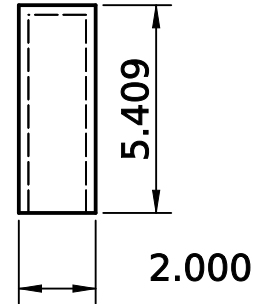
Material 0.500" plate A36		Qty 2	Scale 1:1	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Lid Hinge Lug CINVA-Ram, 14x7x4 brick	Identification number CR-14x7x4-11			Sheet 15/35
		Rev A	Date of issue 2026-09-30		



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. RETAIN WITH WASHER AND $1/8$ COTTER PIN EACH END.

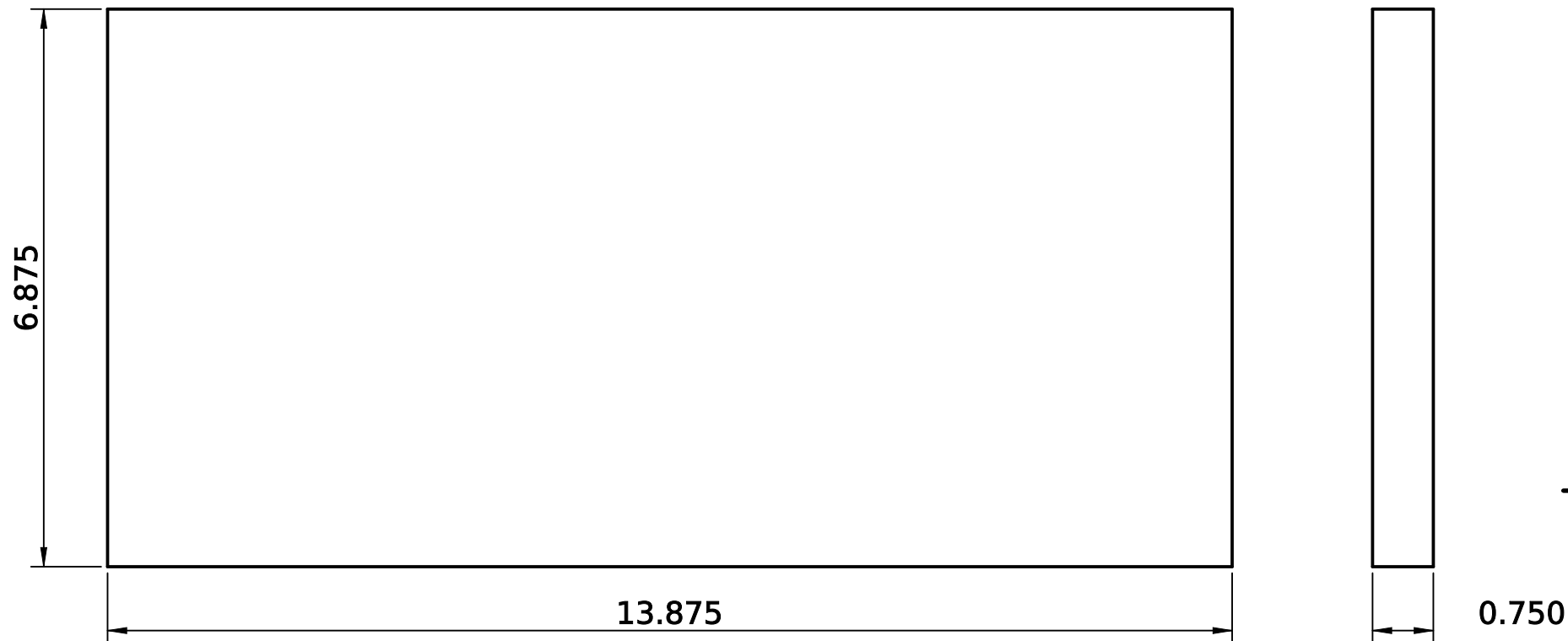
Material 0.750" round 1018		Qty 2	Scale 2:1	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Hinge Pin CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-12		Sheet 16/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. WELD ON THE LID CENTRELINE, CENTRE 1.625 FROM THE LID END AWAY FROM THE HINGE.
5. HEIGHT SETS THE LOCK: THE HANDLE MUST LAND ON IT 3 DEG PAST HORIZONTAL.

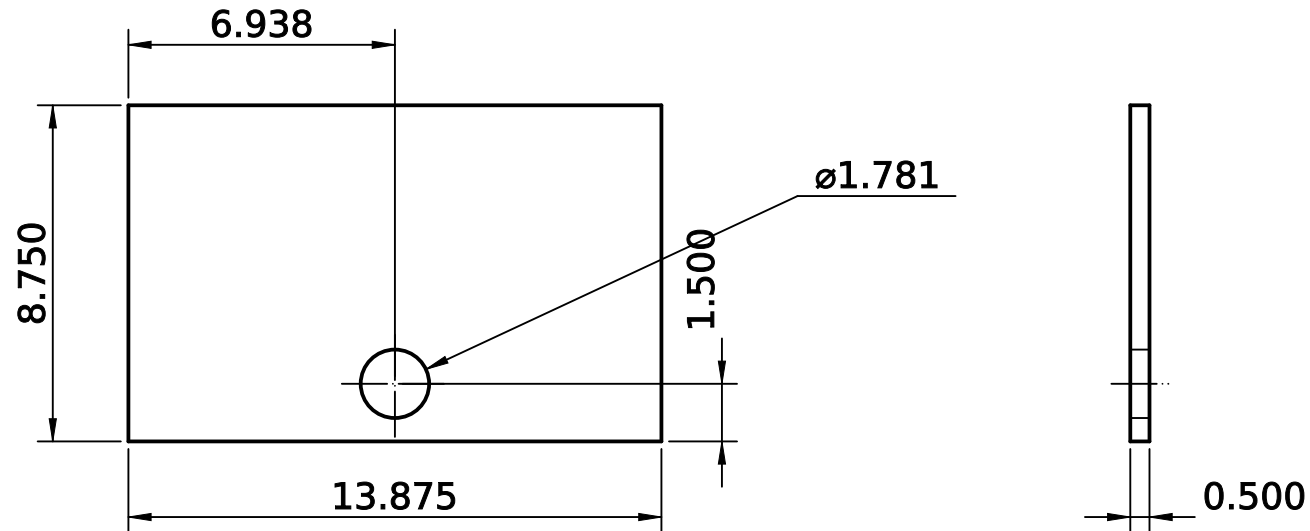
Material 2x2x1/4 sq. tube + cap		Qty 1	Scale 1:5	Units inch	Projection 	
Created by J. Clements III		Approved by -	Document type Detail drawing		Status Draft	
Legal owner J. Clements III CEB Project	Title Handle Stop CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-13		Sheet 17/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. TOP FACE FLAT; BREAK EDGES LIGHTLY. $1/16$ CLEARANCE EACH SIDE IN THE MOLD.

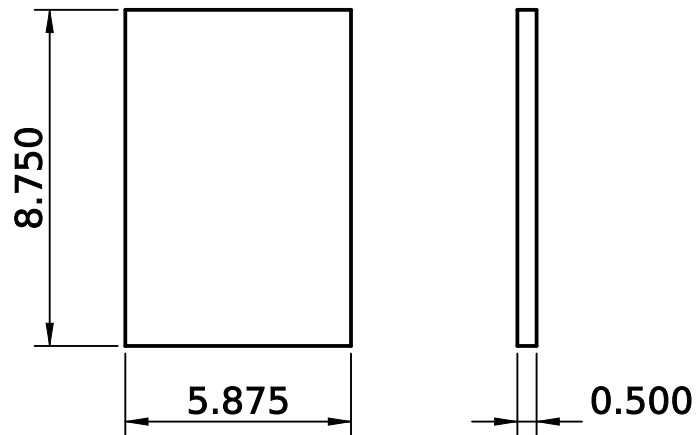
Material 0.750" plate A36		Qty 1	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Piston Cap CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-14	Sheet 18/35
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. WELD UNDER THE CAP FLUSH WITH ITS LONG EDGES; LINE-BORE HOLES AFTER WELDING IF POSSIBLE.

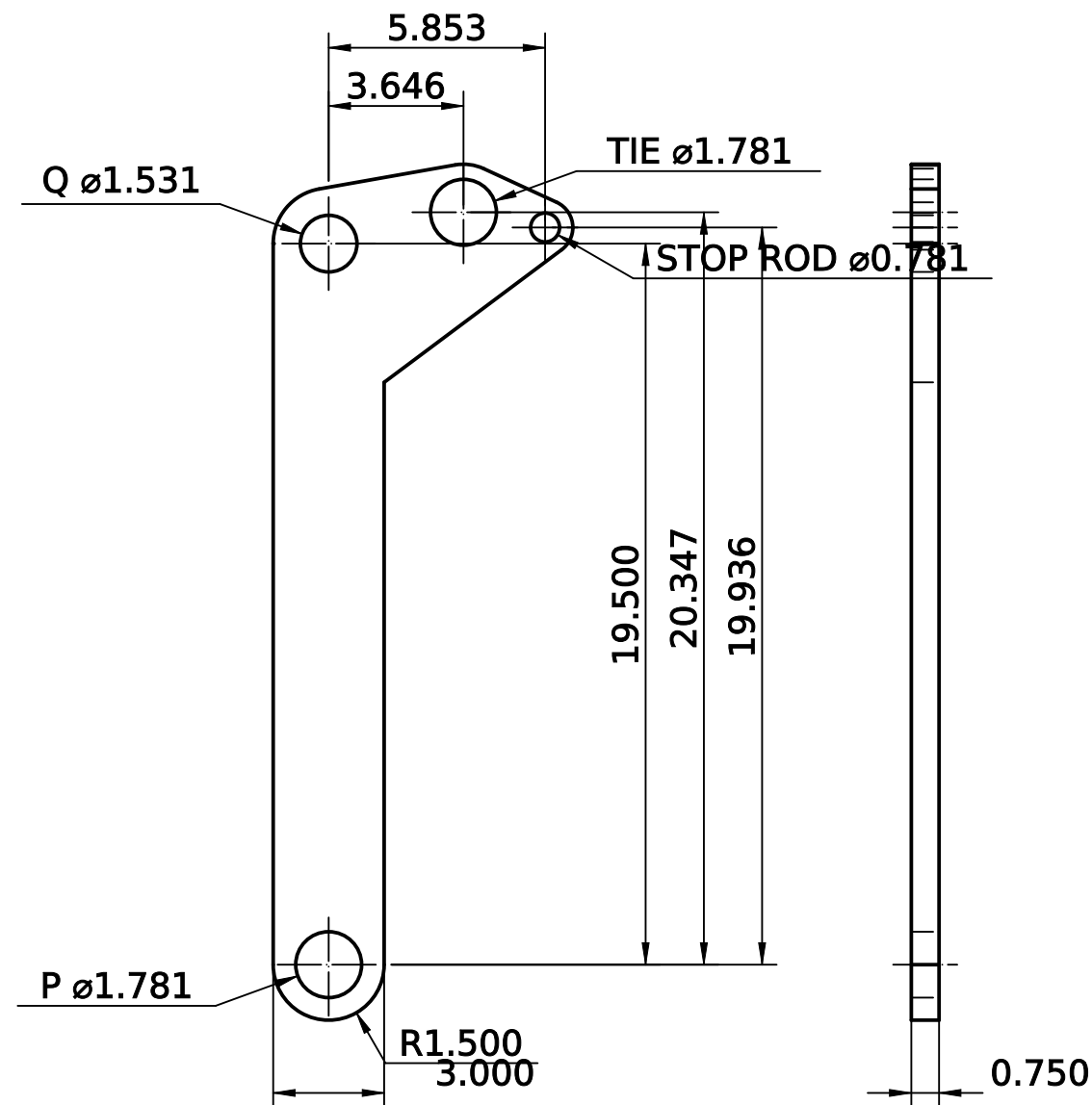
Material 0.500" plate A36		Qty 2	Scale 1:5	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Piston Web CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-15		Sheet 19/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. WELD BETWEEN THE WEBS AT EACH END, UNDER THE CAP.

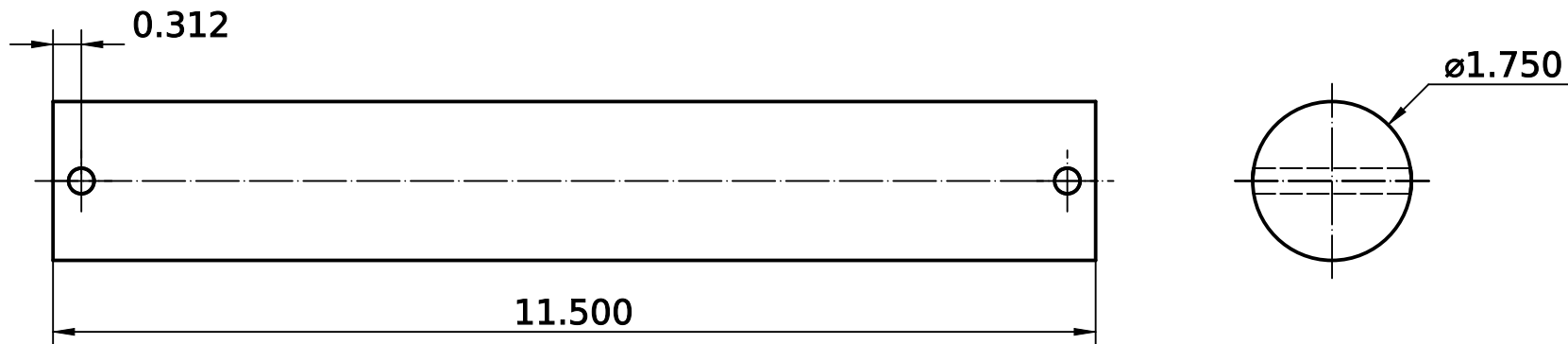
Material 0.500" plate A36		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Piston Diaphragm CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-16	Sheet 20/35
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. DRILL BOTH BARS CLAMPED TOGETHER SO HOLE SPACING MATCHES EXACTLY.
5. HORN OUTLINE: TANGENT ARCS R1.500 ABOUT Q AND 3.000 BELOW IT, R1.300 ABOUT THE TIE HOLE, R0.750 ABOUT THE STOP-ROD HOLE (CUT FROM THE DXF).
6. CLAW TIE AND STOP ROD ARE WELDED INTO BOTH BARS, WHICH MAKES THE YOKE ONE RIGID FRAME.

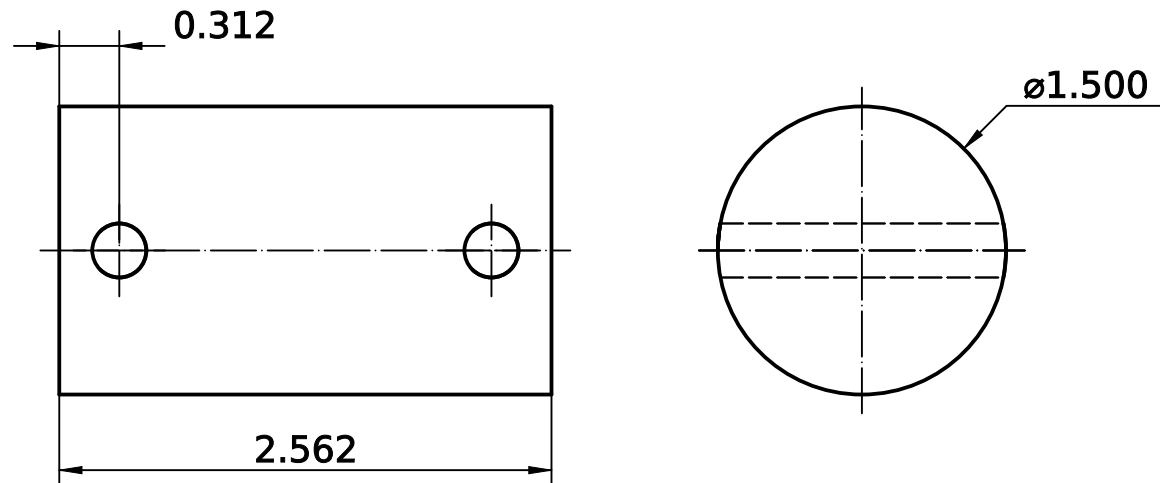
Material 0.750" plate A36 (profile)		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	
Legal owner J. Clements III CEB Project		Title Yoke Bar CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-17
		Rev A		Date of issue 2026-09-30	Sheet 21/35



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. CROSS HOLES $9/32$ FOR $1/4$ HITCH OR COTTER PINS.

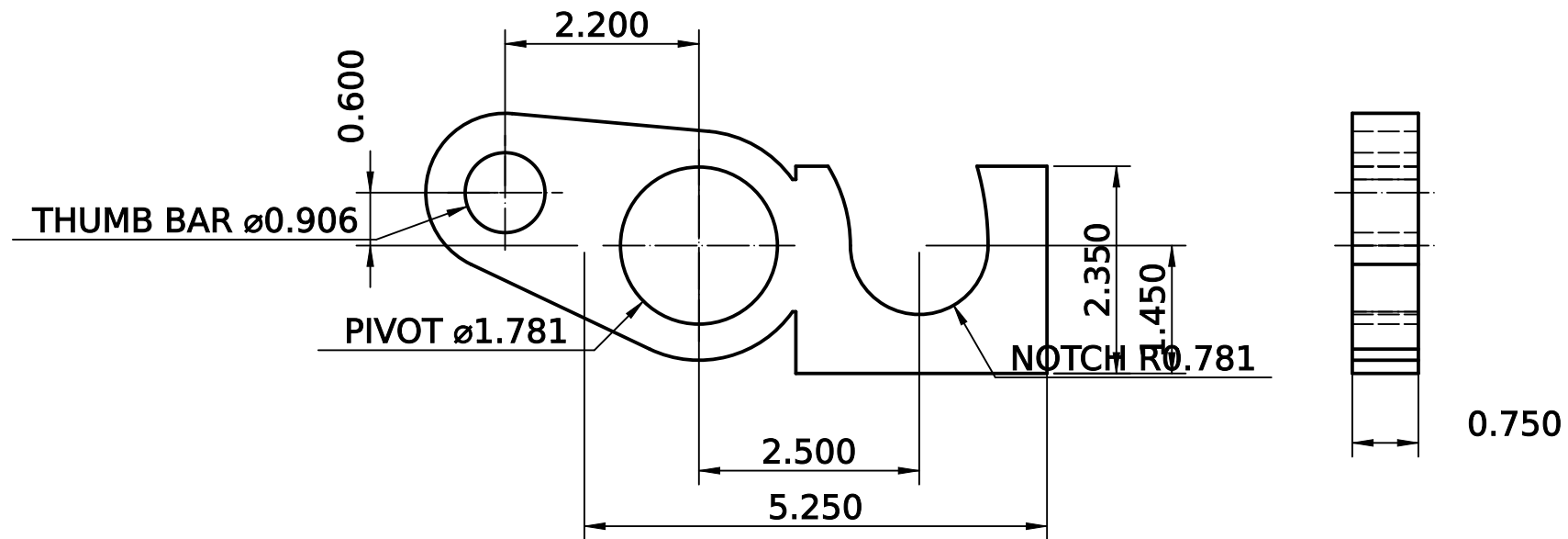
Material 1.750" round 4140 prehard		Qty 1	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Main Pin P CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-18		Sheet 22/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. CROSS HOLES $9/32$ FOR $1/4$ HITCH OR COTTER PINS.

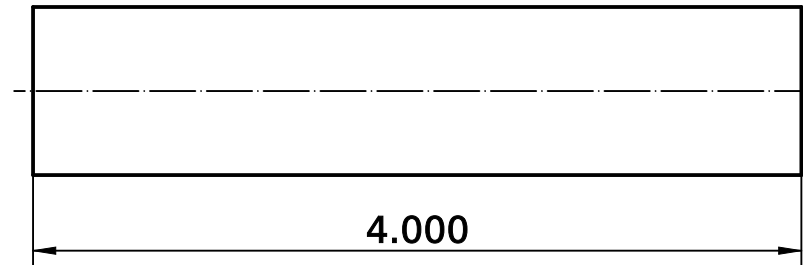
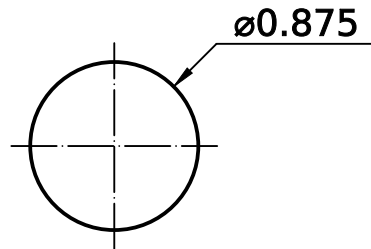
Material 1.500" round 4140 prehard	Qty 2	Scale 1:1	Units inch	Projection 
Created by J. Clements III	Approved by -	Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Stub Pin Q CINVA-Ram, 14x7x4 brick		Identification number CR-14x7x4-19	Sheet 23/35
		Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. TWO IDENTICAL PLATES, WELDED TO THE THUMB BAR 2.500 APART (INSIDE FACES), MAKE ONE CLAW.
5. NOTCH SIDE WALLS ARE ARCS CENTRED ON THE PIVOT HOLE (CUT FROM THE DXF); BREAK ALL EDGES.
6. THE TAIL BEHIND THE PIVOT IS A COUNTERWEIGHT: IT KEEPS THE CLAW SHUT UNDER ITS OWN WEIGHT.

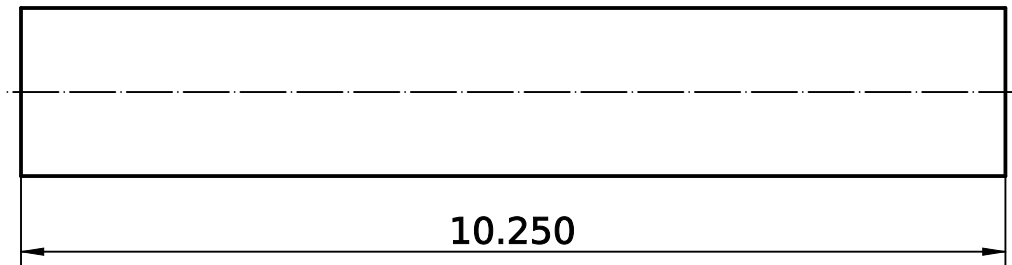
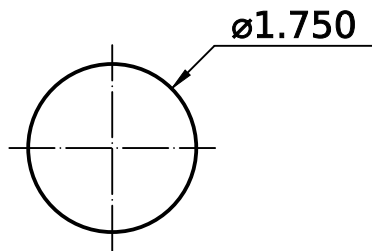
Material 0.750" plate A36		Qty 2	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project		Title Claw Plate CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-20
		Rev A		Date of issue 2026-09-30	Sheet 24/35



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THROUGH BOTH CLAW PLATES, FLUSH WITH THEIR OUTSIDE FACES; WELD BOTH ENDS.
5. PUSH IT DOWN TO FLIP THE CLAW OPEN ONTO THE STOP ROD; LIFT IT TO LET THE CLAW LATCH.

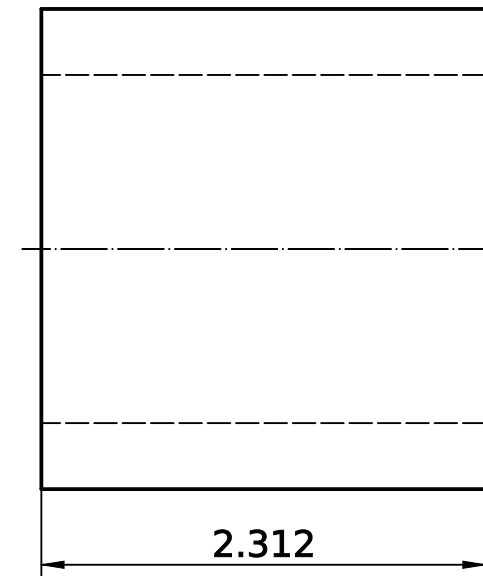
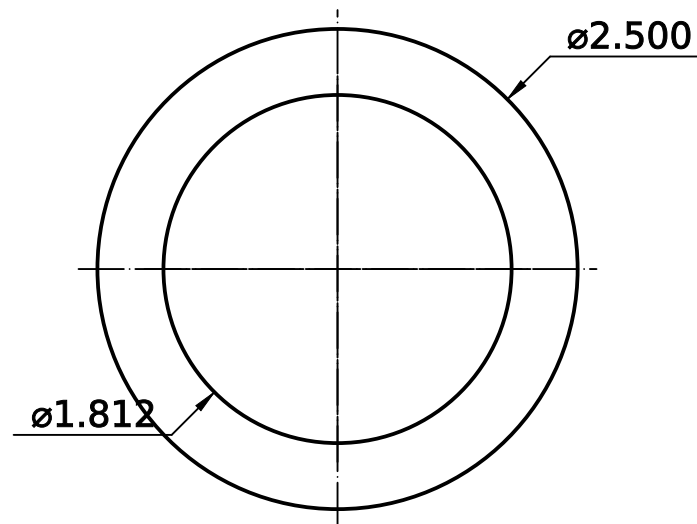
Material 0.875" round 1018		Qty 1	Scale 1:1	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Claw Thumb Bar CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-21		Sheet 25/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THROUGH BOTH YOKE-BAR HORNS, FLUSH OUTSIDE; WELD BOTH ENDS (SMALL WELDS, SLOW COOL).
5. CLAW AND TIE SPACERS GO ON BEFORE THE SECOND END IS WELDED.

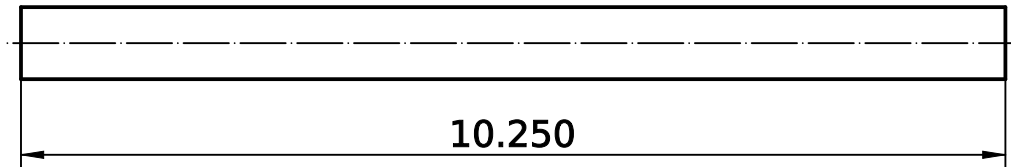
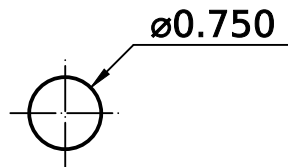
Material 1.750" round 4140 prehard		Qty 1	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Claw Tie CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-22	Sheet 26/35
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. LOOSE ON THE TIE; CENTRES THE CLAW BETWEEN THE YOKE BARS.

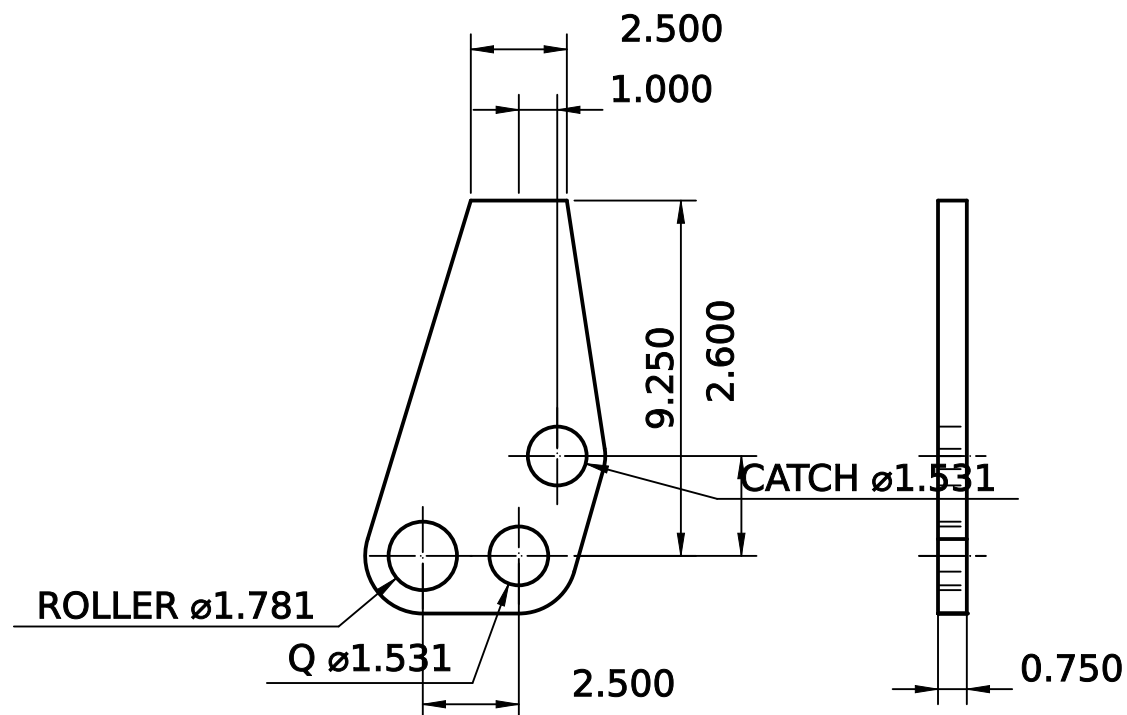
Material 2.5" OD round 1018		Qty 2	Scale 1:1	Units inch	Projection 
Created by J. Clements III		Approved by -	Document type Detail drawing		Status Draft
Legal owner J. Clements III CEB Project	Title Tie Spacer CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-23	Sheet 27/35
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THROUGH BOTH YOKE-BAR HORNS, FLUSH OUTSIDE; WELD BOTH ENDS. THE OPEN CLAW RESTS ON IT.

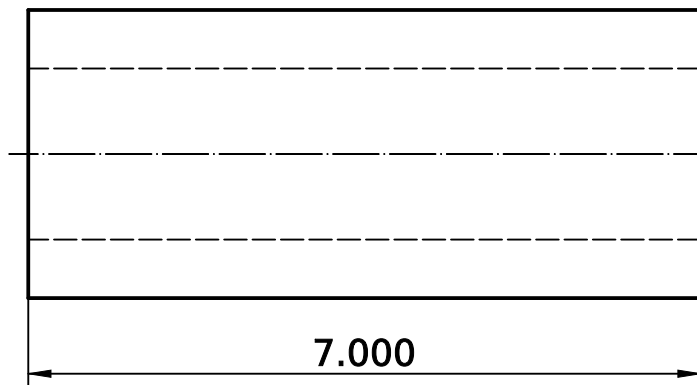
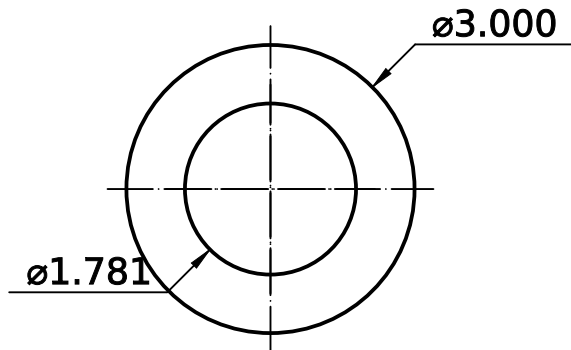
Material 0.750" round 1018		Qty 1	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Claw Stop Rod CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-24		Sheet 28/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. OUTLINE IS TANGENT ARCS R1.500 ABOUT Q AND ROLLER HOLES, R1.250 ABOUT THE CATCH-BAR HOLE.
5. MAKE AS A MIRRORED PAIR; DRILL BOTH CLAMPED TOGETHER.

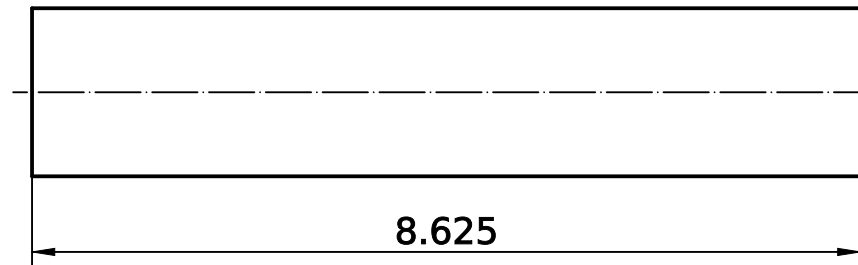
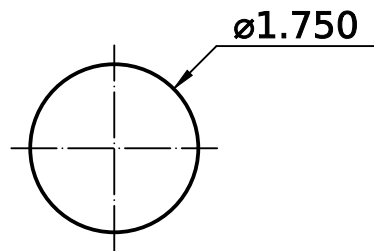
Material 0.750" plate A36		Qty 2	Scale 1:5	Units inch	Projection 
Created by J. Clements III		Approved by -	Document type Detail drawing		Status Draft
Legal owner J. Clements III CEB Project	Title Handle Cheek CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-25	Sheet 29/35
			Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. BORE REAMED FOR A RUNNING FIT ON THE AXLE; GREASE.

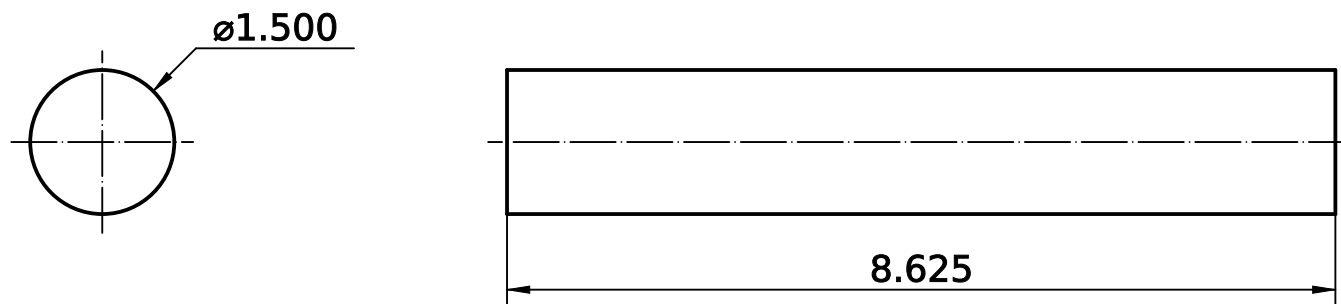
Material 3" OD 4140, 40-45 HRC		Qty 1	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Saddle Roller CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-26	Sheet 30/35
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. WELD AXLE ENDS TO BOTH CHEEKS AFTER THE ROLLER IS INSTALLED.

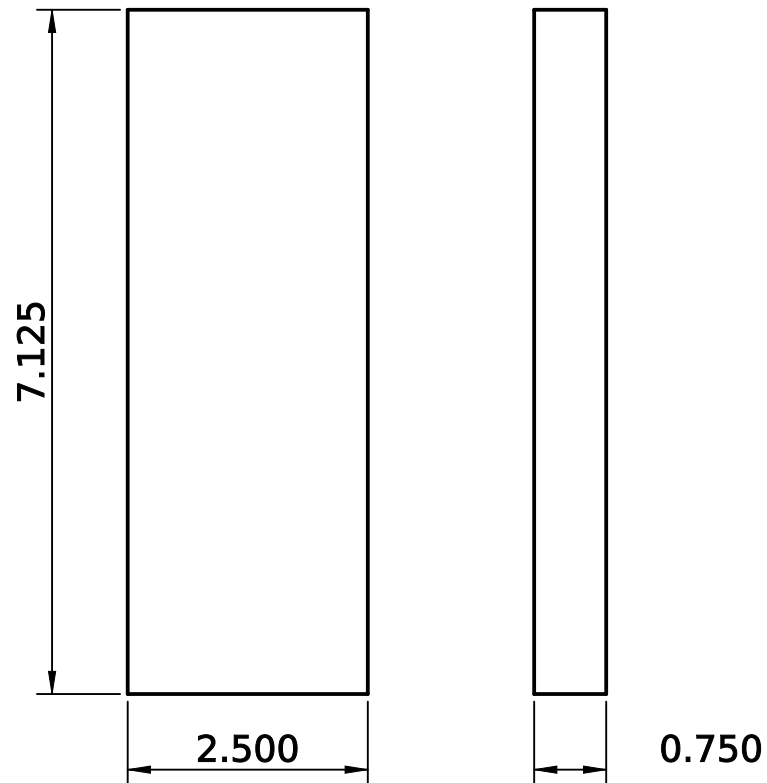
Material 1.750" round 4140 prehard		Qty 1	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Roller Axle CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-27		Sheet 31/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. THROUGH BOTH CHEEKS, FLUSH OUTSIDE; WELD TO EACH CHEEK (SMALL WELDS, SLOW COOL).
5. THE CLAW HOOKS IT IN THE MIDDLE.

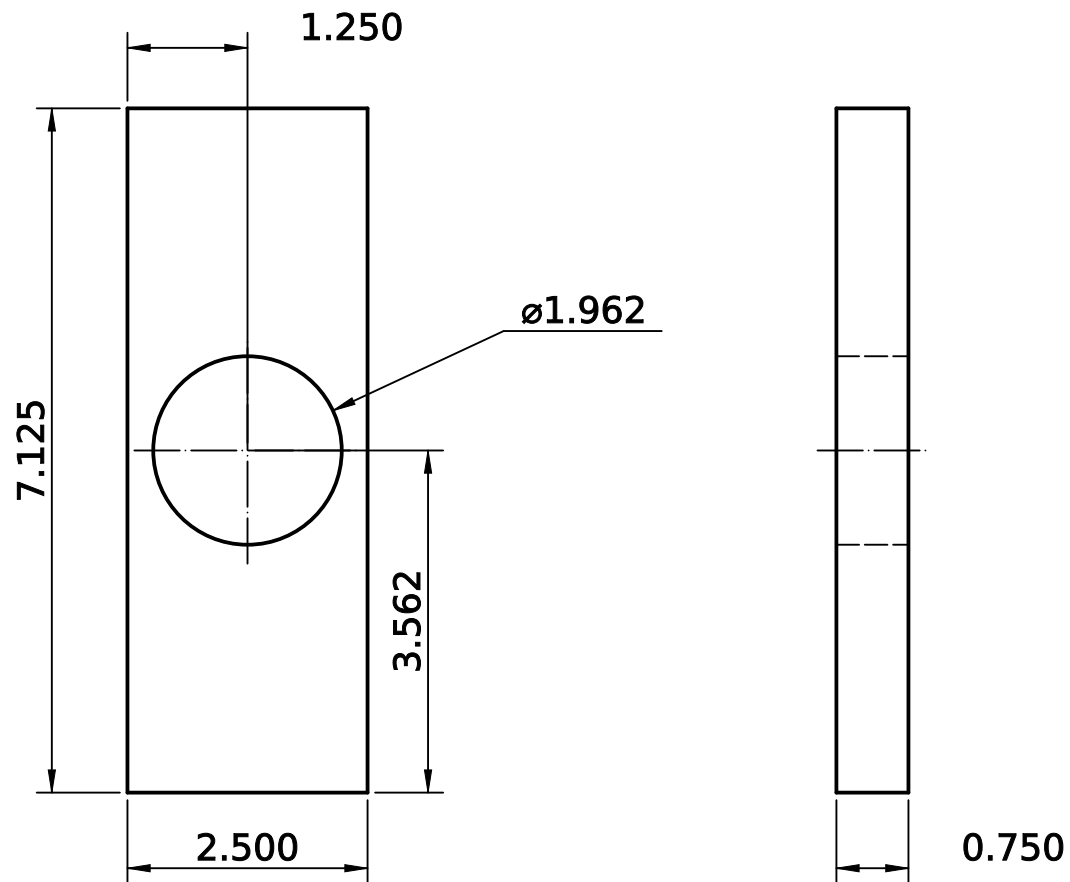
Material 1.500" round 4140 prehard		Qty 1	Scale 1:2	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Catch Bar CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-28	Sheet 32/35
				Rev A	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. WELD BETWEEN THE CHEEKS, 5.750 FROM Q ALONG THE HANDLE LINE.

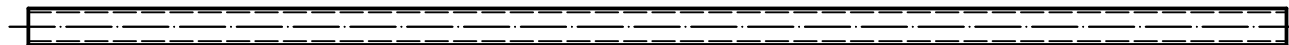
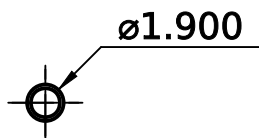
Material 0.750" plate A36		Qty 1	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Handle Bridge CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-29		Sheet 33/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS 1/4 UNLESS NOTED.
4. WELD BETWEEN THE CHEEKS, 8.500 FROM Q ALONG THE HANDLE LINE.

Material 0.750" plate A36		Qty 1	Scale 1:2	Units inch	Projection 	
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft	
Legal owner J. Clements III CEB Project	Title Handle Bridge, Bored CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-30		Sheet 34/35
				Rev A	Date of issue 2026-09-30	



NOTES:

1. ALL DIMENSIONS IN INCHES. TOLERANCE $\pm 1/32$ UNLESS NOTED.
2. BREAK SHARP EDGES, REMOVE BURRS.
3. MATERIAL A36 / 1018 STEEL. FILLET WELDS $1/4$ UNLESS NOTED.
4. BUTT-WELD TO THE PLAIN BRIDGE, FILLET-WELD THROUGH THE BORED BRIDGE.

Material 1-1/2" sch 80 pipe		Qty 1	Scale 1:10	Units inch	Projection 
Created by J. Clements III		Approved by -		Document type Detail drawing	Status Draft
Legal owner J. Clements III CEB Project	Title Handle Tube CINVA-Ram, 14x7x4 brick			Identification number CR-14x7x4-31	Sheet 35/35
				Rev A	